

OpenClaw RGB Strip Control Tutorial

Before starting, please complete the peripheral hardware configuration, API Key configuration, model configuration, and three dialogue solution configuration tests in the "Usage Preparation" directory. If you are currently only using TUI / WhatsApp, you can skip the voice interaction configuration. This article assumes the Jetson Nano Docker environment is already prepared and only covers RGB strip-related content.

1. Tutorial Objectives

This article demonstrates the natural language control capability of OpenClaw on Jetson Nano through the RGB strip peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach for OpenClaw on Jetson to control RGB strips
- Control RGB strips via WhatsApp, TUI, and voice interaction
- Complete cases including solid color lighting, breathing light, warning flashing, running light, horse racing light, and rainbow light
- Understand the complete chain from "user input" to "light execution"

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw Expansion Board
- RGB Strip
- [Optional] AI Voice Interaction Module + Speaker

3. Software Preparation

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Jetson-side entry points used in this tutorial:

```
/root/.openclaw/openclaw.json  
/root/.openclaw/workspace/skills/jetson-hardware-control/SKILL.md  
Corresponding atomic tools (servo/rgb/dht/oled/camera/music)  
/root/yahboom_ws/src/largemodel/main.py
```

Common RGB commands include:

```
/usr/local/bin/rgb --format json color --color red #All turn red  
/usr/local/bin/rgb --format json effect --effect breathe --color blue --speed 5 #Blue  
breathing light  
/usr/local/bin/rgb --format json effect --effect rainbow #Rainbow effect
```

Brightness commands:

```
/usr/local/bin/rgb --format json color --color red --brightness 10  
/usr/local/bin/rgb --format json color --color green --brightness 100
```

4. Testing the RGB Strip

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check. After confirming OpenClaw responds normally, proceed with RGB-specific testing.

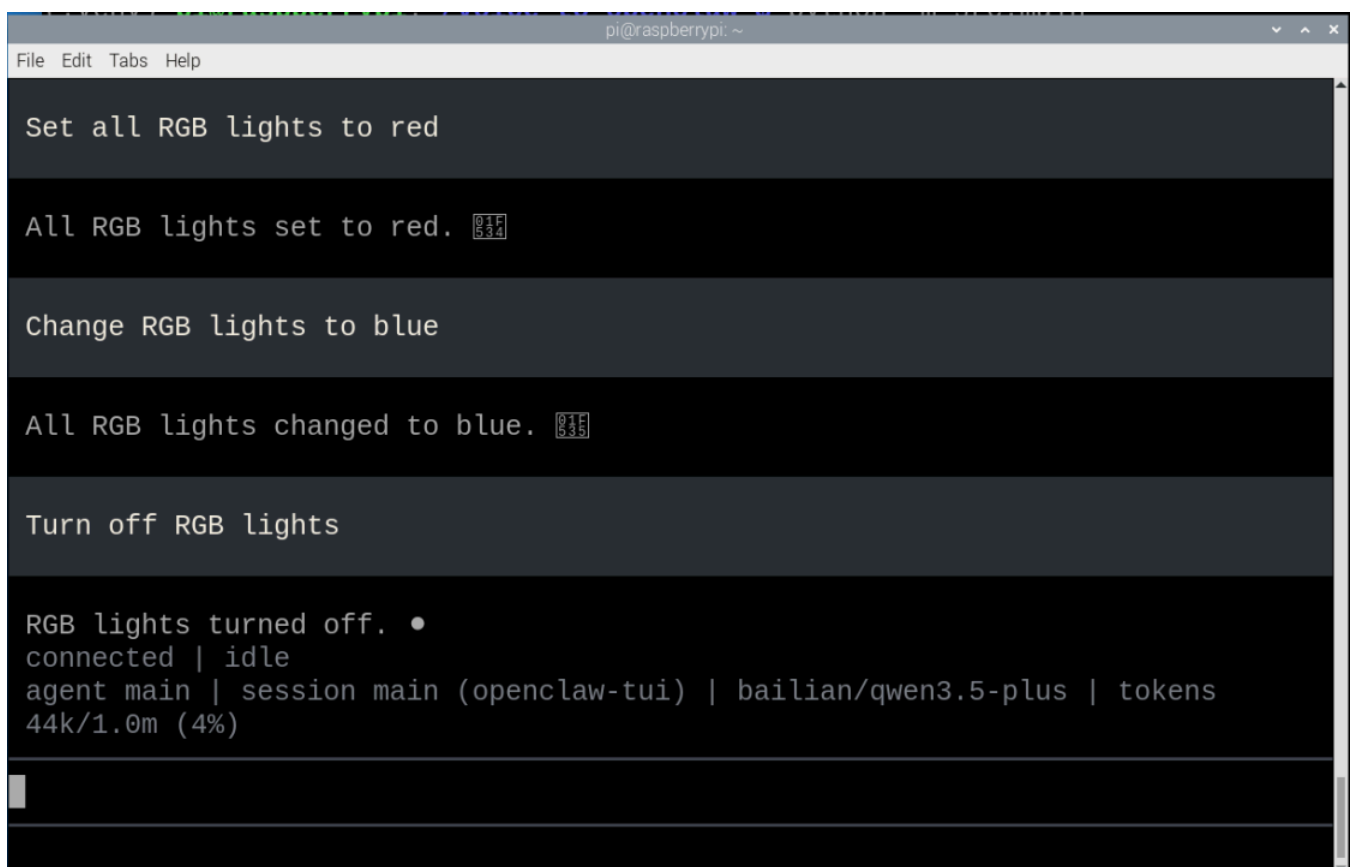
Enter TUI by typing the following command in the terminal:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui  
🔧 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.  
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
session agent:main:main  
gateway connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)  
█
```

In the terminal, you can try inputting the following:

- Set all RGB lights to red
- Change RGB lights to blue
- Lower the RGB light brightness
- Turn off RGB lights



```
pi@raspberrypi: ~  
File Edit Tabs Help  
Set all RGB lights to red  
All RGB lights set to red. 01F  
534  
Change RGB lights to blue  
All RGB lights changed to blue. 01F  
534  
Turn off RGB lights  
RGB lights turned off. •  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens  
44k/1.0m (4%)  
█
```

5. Three Methods for Dialogue with OpenClaw

Method	Suitable Scenario	Advantages	Recommended Use Case
WhatsApp	Remote control, classroom demonstration	Convenient on mobile, suitable for multi-person experience	Demonstrate remote natural language control
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First establish the control link
Voice Interaction	Voice wake-up, complete AI experience	Closest to real dialogue experience	Final demonstration solution

5.1 WhatsApp Solution

For WhatsApp access, QR code creation, and gateway startup steps, please refer to "Three Dialogue Solution Configuration Tests". This section only includes dialogue examples suitable for the RGB tutorial.

You can directly have conversations, for example:

- Make the light strip blue
- Turn on rainbow light effect
- Make the lights flash faster

19:44



< 2



+86 153 3885 7526 (你)

给自己发消息

The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.

19:42 ✓✓

Done! The servo completed a left-to-right swing:

- Moved to 0° (left)
- Swung to 180° (right)
- Returned to 90° (center)

The servo is now ready for the next command.

19:42 ✓✓

Make the light strip blue 19:44 ✓✓

Done! The light strip is now blue at 50% brightness. All 14 LEDs are lit with a nice blue glow.

19:44 ✓✓



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123



space

return



If WhatsApp has text replies but the strip doesn't move, prioritize check if the reply actually triggered `the` corresponding atomic tools (`servo/rgb/dht/oled/camera/music`), and do not directly conclude hardware damage.

5.2 TUI Solution

If TUI self-check is already complete, directly type in the container:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🦖 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

It is recommended to first test fixed colors, then test effects:

- RGB lights turn red
- Set RGB light brightness to 70%
- Turn on blue breathing light

5.3 Voice Interaction Solution

Requires optional AI Voice Interaction Module and speaker.

Execute the following command with openclaw gateway running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

Suitable for directly demonstrating RGB voice commands, for example:

- Take the lights red
- Turn on blue breathing light
- Make the RGB lights into a running light
- Turn on rainbow lights
- Turn off the strip

```
A 2026-05-14 14:56:02 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
C [系统] 串口唤醒已启用: /dev/ttyUSB0@115200
A [提示] 说“清空上下文”可以重置当前会话
T [唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 3780 ms 语音, 正在转写

[你] Take the lights red.
R [发送] 正在交给 OpenClaw
C [完成] OpenClaw 返回 200
a
4 [OpenClaw] Done! The RGB light strip is now red. 🟢

All 14 LEDs are glowing red
at 50% brightness.
```

If the voice has recognized the text but RGB has no action, first check whether `openclaw_bridge` has forwarded the voice text to OpenClaw, and whether OpenClaw has generated an RGB action.

6. Comprehensive Cases

The following cases can be executed in TUI, WhatsApp, or voice. It is recommended to use TUI first during the debugging phase.

6.1 Case 1: Solid Color Lighting

Experiment Objective

Make the RGB lights steadily display a fixed color to complete the most basic light control.

Effect Description

After the RGB lights receive the command, the entire strip uniformly displays the specified color, such as red, green, blue, or yellow. This case is suitable for verifying if color mapping is normal.

Example Commands

- Set all RGB lights to red
- Change the strip to blue
- Adjust the lights to white

6.2 Case 2: Breathing Light

Experiment Objective

Make the RGB lights fade in and out on a certain color to create a breathing light effect.

Effect Description

The RGB lights will change continuously from dark to bright and from bright to dark in a cycle. The overall effect is relatively soft, suitable for standby prompts, ambient lights, or status indicator lights.

Example Commands

- Turn on blue breathing light
- Make the lights into green breathing effect
- Set RGB to red breathing light

6.3 Case 3: Warning Flashing

Experiment Objective

Make the RGB lights flash to simulate warning lights or alarm effects.

Effect Description

Lights can flash using high-contrast colors such as red and yellow, suitable for expressing "attention", "warning", "abnormal status", and other information.

Example Commands

- Make the lights flash red
- Turn on yellow warning lights
- Make the RGB lights flash quickly

6.4 Case 4: Running Light and Horse Racing Light

Experiment Objective

Make the RGB lights create continuous flowing dynamic effects to demonstrate richer light performances.

Effect Description

Both running lights and horse racing lights can show dynamic light effects, suitable for boot animations, prompt animations, or interactive displays.

Example Commands

- Turn on rainbow effect
- Turn on blue running light
- Turn on yellow horse racing light
- Make RGB lights into rainbow lights

7. Common Problems

- RGB has no action: First execute `rgb color --color red` to confirm if the atomic tool is functioning normally.
- Hardware commands can get command hints via `-h`:

```

root@jetson-yahboom:~# rgb -h
usage: rgb [-h] [--dry-run] [--format {json,text}] {color,effect,off} ...

RGB 灯条控制 - 原子操作 + 特效灯

positional arguments:
  {color,effect,off}
    color              设置颜色
    effect             设置特效灯
    off               关闭灯条

options:
  -h, --help            show this help message and exit
  --dry-run            模拟执行, 不实际控制硬件
  --format {json,text} 输出格式
root@jetson-yahboom:~# rgb color -h
usage: rgb color [-h] [--index INDEX] [--color COLOR] [--r R] [--g G] [--b B] [--brightness BRIGHTNESS]

options:
  -h, --help            show this help message and exit
  --index INDEX        灯珠索引 (0-13 或 255 表示全部)
  --color COLOR        颜色值 (#RRGGBB 或颜色名称)
  --r R                红色通道 (0-255)
  --g G                绿色通道 (0-255)
  --b B                蓝色通道 (0-255)
  --brightness BRIGHTNESS 亮度 (0-100)
root@jetson-yahboom:~# rgb color --color red
{"ok": true, "code": "ok", "message": "RGB 颜色已设置", "details": {"index": 255, "r": 255, "g": 0, "b": 0}}
root@jetson-yahboom:~# rgb color --color red -h
usage: rgb color [-h] [--index INDEX] [--color COLOR] [--r R] [--g G] [--b B] [--brightness BRIGHTNESS]

options:
  -h, --help            show this help message and exit
  --index INDEX        灯珠索引 (0-13 或 255 表示全部)
  --color COLOR        颜色值 (#RRGGBB 或颜色名称)
  --r R                红色通道 (0-255)
  --g G                绿色通道 (0-255)
  --b B                蓝色通道 (0-255)
  --brightness BRIGHTNESS 亮度 (0-100)
root@jetson-yahboom:~# rgb color --color red --brightness 10
{"ok": true, "code": "ok", "message": "RGB 颜色已设置", "details": {"index": 255, "r": 26, "g": 0, "b": 0}}
root@jetson-yahboom:~# █

```